CS2705 — Programming and Data Structures

Tier	A
Phase (order)	2
Do this first	CS1111
Estimated effort	10-12 weeks, ~8 h/week
Unlocks	CS2800, CS3500, CS3700, CS3300

Roadmap

1. Asymptotics (big-oh/omega/theta) and how to count steps.
2. Arrays: searching, then sorting (bubble, insertion, merge, counting, radix, bucket).
3. Linked lists, stacks, queues. Implement each from scratch in C.
4. Binary trees, traversals, BST, AVL; hashing and collision handling; binary heaps and priority queues.
5. Graph basics: adjacency list/matrix, BFS, DFS.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Computer Sc - Programming and Data Structure (32 videos). NPTEL *Programming and Data Structure* (nptelhrd): C-based PDS, matches IITM CS2705's C focus.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Programming, Data Structures And Algorithms Using Python (45 videos). NPTEL *Programming, Data Structures and Algorithms using Python* (Madhavan Mukund, CMI): the clearest PDS teacher in India.
- MIT 6.006 Introduction to Algorithms, Spring 2020 (32 videos). MIT 6.006 *Introduction to Algorithms* (Spring 2020): data structures + algorithms, rigorous.

⚡ Short videos for intuition and revision

- Algorithms (84 videos). Abdul Bari *Algorithms*: whiteboard explanations, perfect when a topic doesn't click.
- Data structures playlist (55 videos). WilliamFiset *Data structures*: animated, with code.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–4	C refresher at PDS speed. Re-do pointer exercises from CS1111.

Week	Lectures	What to do
2	5–11	Data structuring case studies, recursion, mergesort/quicksort. Implement both sorts in C.
3	12–17	Strings, arrays/addresses, structures, dynamic allocation, linked lists. Implement a singly + doubly linked list.
4	18–21	Complexity and asymptotic analysis. Time your sorts on 10^3..10^6 elements and plot.
5	22–25	Search trees, 2-3 trees. Implement a BST with insert/delete/inorder.
6	26–32	Algorithm design + graphs. Implement BFS/DFS on an adjacency list. Then stacks/queues/heaps/hashing from Mukund's Python course (alt playlist).

▼ **All lectures in *Computer Sc - Programming and Data Structure* (32 videos)**

1. Lec-1 Introduction
2. Lec-2 C Programming-I
3. Lec-3 C Programming-II
4. Lec-4 C Programming-III
5. Lec-5 Data Structuring: Case Study-I
6. Lec-6 Data Structuring: Case Study-II
7. Lec-7 Data Structuring: Case Study-III
8. Lec-8 Problem Decomposition By Recursion-I
9. Lec-9 Problem Decomposition By Recursion-II
10. Lec-10 Problem Decomposition By Recursion-III
11. Lec-11 Mergesort And Quicksort
12. Lec-12 Characters And Strings
13. Lec-13 Arrays: Addresses And Contents
14. Lec-14 Structures-I
15. Lec-15 Structures-II
16. Lec-16 Dynamic Allocation Part-I
17. Lec-17 Linked Lists-I
18. Lec-18 Complexity(Efficiency) of Algorithms
19. Lec-19 Asymptotic Growth Functions
20. Lec-20 Asymptotic Analysis of Algorithms
21. Lec-21 Data Structuring
22. Lec-22 Search Trees
23. Lec-23 Search Trees-II
24. Lec-24 Search Trees-III
25. Lec-25 2-3 Trees
26. Lec-26 Algorithm Design-I
27. Lec-27 Algorithm Design-II

28. Lec-28 Algorithm Design-III
29. Lec-29 Graphs-I
30. Lec-30 Graphs-II
31. Lec-31 Graphs-III
32. Lec-32 Conclusions

Books, notes and websites

- ✅ NPTEL *Programming and Data Structures* (N.S. Narayanaswamy, IIT Madras) — nptel.ac.in/courses/106106130/
- ✅ NPTEL *Programming, DSA using Python* (Madhavan Mukund, CMI) — if you prefer Python
- Princeton *Algorithms, Part I* (Sedgewick & Wayne) — algs4.cs.princeton.edu (site was unreachable from my check; try again, or use Coursera)
- Book: *Data Structures and Algorithm Analysis in C++* (Weiss)
- Practice: LeetCode Easy->Medium, 3 problems/week by topic

Practice / proof of understanding

Implement linked list, stack, queue, BST, AVL, hash table, binary heap and BFS/DFS from scratch, each with a test file.

Note

Same ground as DA1300/DA5300 in the DA department, taught in C here.

What your IITM course sheet says

From the PDF: CS2705

Official description: The objective of the course is to teach programming with an emphasis on problem solving and introduce elementary data structures. The student should be able to analyze computational efficiency.

Official syllabus (trimmed):

> Review of Problem Solving using computers, Abstraction, Elementary Data Types, Algorithm design (2 lectures) Complexity and Efficiency via model of computation (notion of time and space), mathematical preliminaries, Elementary asymptotics (big-oh, big-omega, and theta notations). (3 lectures) ADT Array -- searching and sorting on arrays:Linear search, binary search on a sorted array. Bubble sort, Insertion sort, Merge Sort and analysis; Emphasis on the comparison based sorting model. Counting sort, Radix sort, bucket sort. (6 lectures) ADT Linked Lists, Stacks, Queues:List manipulation, insertion, deletion, searching a key, reversal of a list, use of recursion to reverse/search. Doubly linked lists and circular linked lists. (4 lectures) Stacks and queues as dynamic data structures implemented using linked lists. Analyse the ADT operations when implemented using arrays. (3 lectures) ADT Binary Trees:Tree representation, traversal, application of binary trees in Huffman coding. Introduction to expression trees: traversal vs post/pre/infix notation. Recursive traversal and other tree parameters (depth, height, number of nodes etc.) (5 lectures) ADT Dictionary: Binary search trees, balanced binary search trees - AVL Trees. Hashing - collisions, open and closed hashing, properties of good hash functions. (3+3 lectures) ADT Priority queues: Binary heaps with applicati

Official textbooks: Data Structures and Algorithm Analysis in C++, by Mark Allen Weiss (Pearson 2007).

Official references: Data structures and Algorithms in C++ -- by Adam Drozdek (1994 2001). How to solve it by Computer -- by R G Dromey (PHI 1982, Paperback 2008). Fundamentals of Data Structures in C -- by Horowitz, Sahni and Anderson-Freed (Silicon Press 2007). Data Structure Using C and C++ -- by Y. Langsam, M. J. Augenstein and A. N. Tanenbaum (Pearson Education, 2nd Edition, 2015).